Quantification: 548_hyp_HP_z08c1-2
==================================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	14
interior lacuna count	13
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 13)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	3146.54	3031	2299.75	px^2
major_axis_length_px	lacuna shape	89.1092	97.3283	37.3865	px
minor_axis_length_px	lacuna shape	46.0008	47.2353	20.8832	px
aspect_ratio	lacuna shape	2.0744	1.7987	0.9019	unitless
eccentricity	lacuna shape	0.8168	0.8312	0.112	unitless
solidity	lacuna shape	0.7998	0.8013	0.0907	unitless
roots_count	headline	11.1538	9	7.0693	count
ring_length_r30_px	headline	341.846	348	171.252	px
ring_length_r60_px	reported	653.769	654	269.239	px
owned_length_px	ownership-dependent	1001.19	971.323	623.799	px
edge_count	ownership-dependent	48.6923	48	30.3929	count
mean_edge_length_px	ownership-dependent	20.3314	19.7353	3.9079	px
perimeter_px	normalised	290.084	274.007	143.253	px
ring_area_r30_px2	normalised	9736.46	10180	3681.94	px^2
ring_area_r60_px2	normalised	21257.3	22542	7120.36	px^2
in_frame_fraction_r30	normalised	0.9801	1	0.0453	unitless
in_frame_fraction_r60	normalised	0.9529	1	0.0937	unitless
ring_density_r30	normalised	0.034256	0.0366405	0.00826687	px^-1
ring_density_r60	normalised	0.0308581	0.0314017	0.00638511	px^-1
roots_per_100px_perimeter	normalised	3.7714	3.9758	1.1517	per 100 px
ring_attached_length_r30_px	network v2	305.538	336	168.796	px
ring_attached_length_r60_px	network v2	466.462	560	265.059	px
ring_length_w_r30_px	network v2	390.376	396.049	195.53	px
ring_length_w_r60_px	network v2	746.894	741.713	306.498	px
sholl_crossings_r10	network v2	8.5385	8	4.2743	count
sholl_crossings_r20	network v2	8.9231	8	4.663	count
sholl_crossings_r30	network v2	8.1538	6	4.4506	count
loop_count	loops, report only	2.4615	2	2.1454	count
loop_edge_share	loops, report only	0.1497	0.125	0.1217	share
ring_width_mean_r30_px	width	7.5075	7.6251	0.6129	px

Per field
----

measure	value	units
analysed_area_px2	999561	px^2
total_skeleton_length_px	26683	px
canalicular_length_density_per_px	0.0266947	px^-1
junction_count	525	count
junction_density_per_px2	0.00052523	px^-2
lacuna_count	14	count

lacunae_per_px2	1.401e-05	px^-2
skeleton_component_count	377	count
mean_component_length_px	70.7772	px
median_component_length_px	21	px
field_length_density_w_per_px	0.030522	px^-1
field_density_without_flagged_per_px	0.0266947	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	277	px
unattached_component_count	16	count
canalicular_length_density_attached_per_px	0.0266947	px^-1
field_length_density_with_unattached_per_px	0.0269718	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	43	count
loop_edge_share	0.106	share
edge_count	1406	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6339	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	26683	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3961	122.826	46.3249	2.6514	0.9261	0.8431	0.4753	53.5506	452.253	False
2	547	38.4822	18.6369	2.0648	0.8749	0.9071	1.3303	28.7715	719.596	False
3	4287	134.045	52.6927	2.5439	0.9195	0.6794	-0.8737	66.8097	864.372	False
4	3031	74.7141	59.7338	1.2508	0.6007	0.7786	-0.0729	198.233	105.492	False
5	1249	97.3283	20.9931	4.6362	0.9765	0.6665	0.811	244.751	234.965	False
6	833	42.3386	25.9256	1.6331	0.7906	0.8976	0.935	290.168	253.631	False
7	8164	134.063	85.066	1.576	0.7729	0.8013	-0.8691	359.846	342.936	False
8	5658	108.902	72.88	1.4943	0.7431	0.7761	-0.0527	409.015	685.903	False
9	2786	87.8405	47.2353	1.8596	0.8431	0.8199	0.6054	469.408	431.352	False
10	4506	98.6377	61.8033	1.596	0.7794	0.8877	-0.0114	577.89	721.96	False
11	534	30.0222	23.0801	1.3008	0.6395	0.9287	0.8021	729.066	431.221	False
12	1086	59.0095	32.8062	1.7987	0.8312	0.7126	1.4627	783.261	210.357	False
13	4263	130.21	50.8324	2.5616	0.9207	0.6991	-1.4214	797.485	870.568	False
14	8110	200.348	57.8345	3.4642	0.9574	0.7865	-1.2162	906.084	938.645	True

Network per lacuna

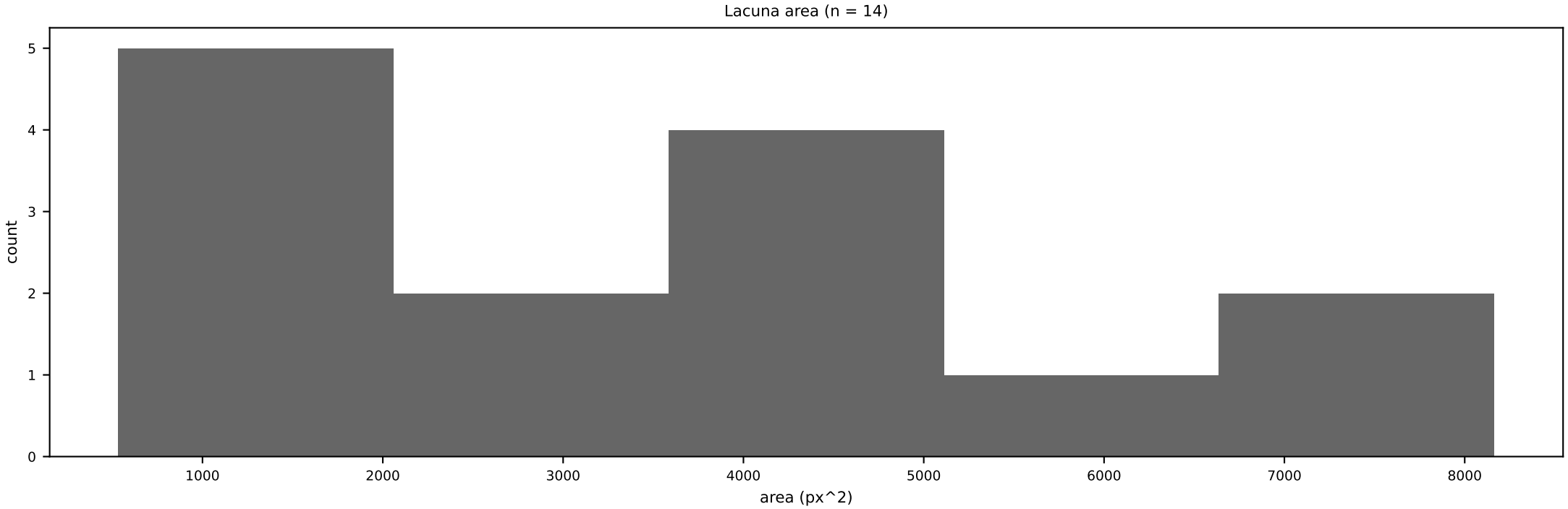
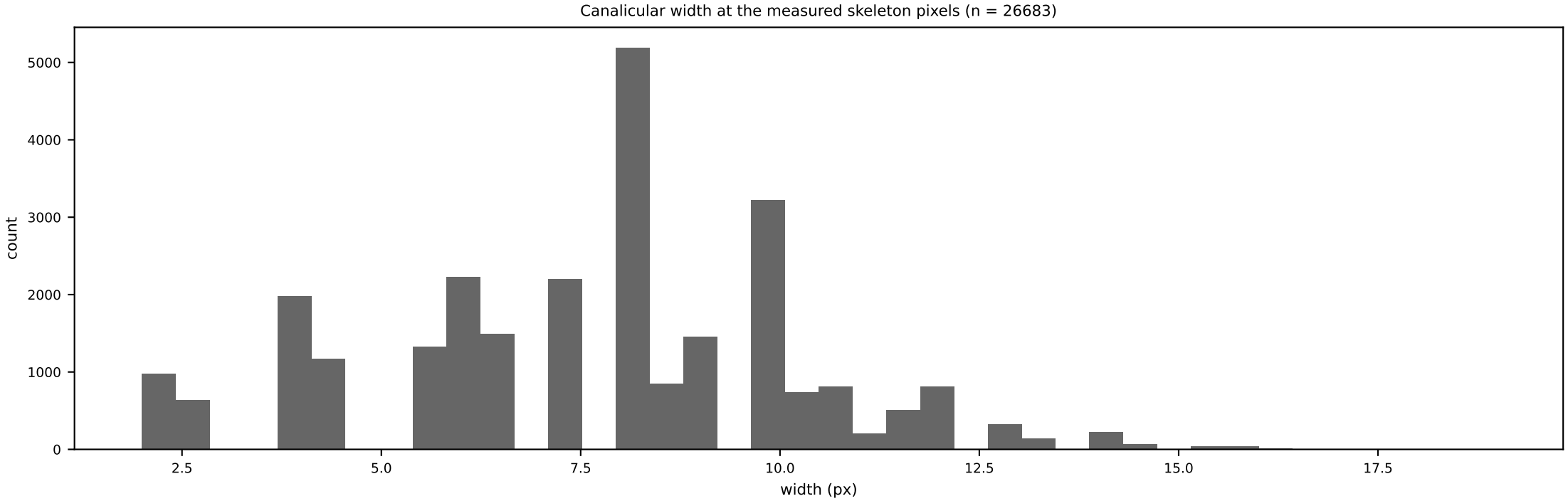
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	251	503	1112.97	68	16.3673	8.3083
2	6	228	482	827.252	28	29.5447	7.1888
3	20	530	898	1829.13	87	21.0244	7.6483

4	6	403	827	1056.68	47	22.4825	7.1285
5	10	278	427	273.794	17	16.1055	7.3603
6	5	137	226	971.323	48	20.2359	8.2717
7	25	686	1221	2131.41	108	19.7353	7.7058
8	20	427	776	952.051	50	19.041	7.6251
9	6	373	723	1361.28	53	25.6846	8.1493
10	13	348	654	1562.55	72	21.7021	7.8628
11	2	99	384	17.0711	1	17.0711	6.0603
12	7	169	493	205.752	11	18.7048	7.2964
13	16	515	885	714.169	43	16.6086	6.9922
14	16	522	855	846.838	41	20.6546	8.3408

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	327.706	10056	22309	0.8579	0.7712	0.0249602	0.0225469	2.7464
2	102.326	5103	11344	0.9041	0.7441	0.0446796	0.0424894	5.8636
3	434.274	13343	26762	0.9794	0.8718	0.0397212	0.033555	4.6054
4	274.007	10299	23608	1	1	0.03913	0.0350305	2.1897
5	225.38	6965	13598	1	1	0.0399139	0.0314017	4.437
6	121.983	3984	7252	1	1	0.0343876	0.0311638	4.0989
7	522.985	15465	29728	1	1	0.0443582	0.0410724	4.7803
8	451.63	13578	29430	1	1	0.0314479	0.0263676	4.4284
9	268.777	10180	22542	1	1	0.0366405	0.0320735	2.2323
10	328.534	11289	25601	1	1	0.0308265	0.0255459	3.957
11	95.6396	5395	16431	1	1	0.0183503	0.0233705	2.0912
12	176.066	7548	20526	1	1	0.02239	0.0240183	3.9758
13	441.789	13369	27214	1	1	0.0385219	0.03252	3.6216
14	574.014	14230	26716	0.8478	0.7913	0.0366831	0.0320033	2.7874

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	206	281	282.794	578.63	6	6	4
2	215	321	258.309	549.931	4	6	3
3	459	741	600.387	1037.22	12	10	15
4	336	704	457.747	935.151	8	14	14
5	220	286	315.25	478.375	6	11	5
6	105	164	163.095	261.522	2	5	4
7	663	997	787.311	1390.21	16	20	12
8	385	560	488.345	891.607	11	9	11
9	348	560	432.718	825.767	14	8	10
10	339	562	396.049	741.713	10	8	13
11	82	107	113.054	443.688	5	2	3
12	127	187	192.51	568.328	5	5	6
13	487	594	587.316	1007.48	12	12	6
14	490	655	592.902	974.791	12	12	17

548_hyp_HP_z08c1-2: distributions, pre-validation, px